

Mahdi Haghverdi

Software Engineer | Cloud Native & DevOps Enthusiast

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Software Engineer with 2 years of professional experience building production backend systems and microservices, backed by over 6 years of Python programming, self-directed learning, and personal projects. Experienced in observability, distributed tracing, containerized deployments, CI/CD, and event-driven architectures. Currently expanding expertise in Kubernetes and cloud-native infrastructure while focusing on reliability, automation, and developer productivity.

SKILLS

- **Tools and Technologies**
 - **Backend:** Python, FastAPI, FastStream, Redis, Redis Streams, Pydantic, PydanticAI, SQLAlchemy
 - **Observability:** OpenTelemetry, Grafana, Grafana Alloy, Grafana Tempo, FluentBit, VictoriaMetrics, VictoriaLogs, LogFire, StructLog
 - **Architecture:** Microservices, REST APIs, Event-driven Architecture, System Design
 - **DevOps & Cloud:** Docker, Docker Compose, Kubernetes (learning), OpenStack (learning)
- **Methodologies and Industry Knowledge:** Agile, Scrum, System Design, Microservices
- **Soft Skills:** Problem Solving, Communication, Collaboration, Adaptability, Ownership, Continuous Learning

EXPERIENCE

Software Developer | Jast, Health Technology Co.

Jun 2024 - PRESENT

Part-time | Remote

- **Monitoring & Logging:**
 - Implemented cloud-native structured logging across all services using Python Logging and Structlog.
 - Separated dev and production log formats to improve local debugging while preserving machine-readable production logs.
 - Configured Fluent Bit sidecar containers to forward structured logs to VictoriaLogs.
 - Integrated Logfire and Grafana Alloy to collect distributed traces across microservices.
 - Implemented distributed tracing across production microservices, reducing incident investigation time by approximately 80%.
 - Configured Grafana Tempo as the distributed trace backend.
 - Collected application metrics using Logfire and Grafana Alloy.
 - Implemented custom metrics for various business requirements.
 - Designed operational dashboards enabling developers to identify failures and performance bottlenecks in real time.
- **Microservices:**
 - Built an event-driven Web-Push API notification service and real-time in-app user notifications.
 - Developed an event-driven service for email and SMS delivery.
 - Implemented an image processing service for compression, resizing, cropping, and video thumbnail generation.
 - Reduced image size by 50% to 90%.
 - Built a real-time chat service enabling communication between trainers and trainees.
- **Technical Work:**
 - Introduced a service layer architecture to separate business logic from API and persistence layers.
 - Set up unit and integration tests using Pytest.
 - Designed and developed new Pydantic schemas for the REST APIs to be more concise and accurate in terms of validation and API documentation.
 - Migrated SQLAlchemy models to the 2.0 declarative syntax with full type annotations.
 - Developed a new Exception hierarchy for better error handling and logging.
 - Introduced domain-specific error codes for consistent API error handling.
 - Improved API usability by introducing standardized error codes and documenting them in OpenAPI (Swagger).
- **CI/CD:**
 - Built GitHub Actions workflows for automated testing and validation.

EDUCATION

Bachelor of Engineering, Computer Engineering | University of Isfahan

2021-2025

Thesis: Design and Implementation of a POSIX-Compatible Filesystem for Direct S3-Based Storage Access

Grade: 19/20

Project: <https://github.com/mahdihaghverdi/pys3fuse>

- Developed transparent FUSE filesystem layer to access S3-based storages with pure POSIX commands (ls, cd, mkdir, ...)
- Implemented a fully type-safe codebase with comprehensive type annotations.
- Implemented comprehensive unit tests using Pytest.
- Implemented concurrent background operations using multithreading and non-blocking design patterns.